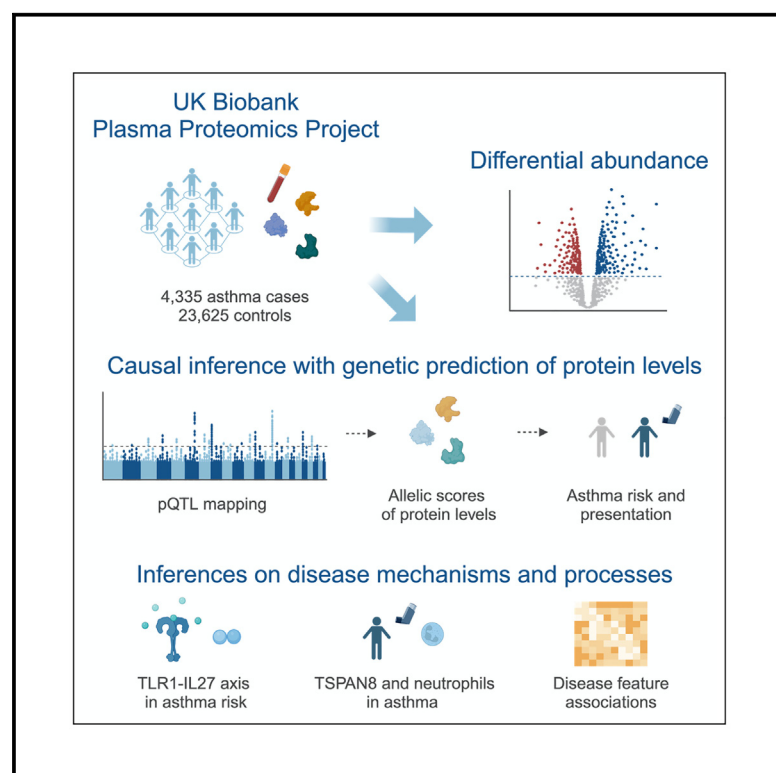


Integration of biobank-scale genetics and plasma proteomics reveals evidence for causal processes in asthma risk and heterogeneity

Graphical abstract



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In brief

Plasma proteomics may capture key insights into disease mechanisms and reveal informative biomarkers. Donoghue et al. apply a causal inference framework to genetic and plasma proteomic data from the UK Biobank and demonstrate that genetically predicted protein levels are effective instruments for identifying putatively causal factors underlying both disease risk and heterogeneity.

Highlights

- Genetically predicted protein levels reveal putatively causal mechanisms in asthma
- Identification of candidate downstream biomarkers of asthma
- Causal protein candidates have differential associations with asthma endophenotypes
- Evidence for a TLR1-IL-27 axis in asthma risk and TSPAN8 in asthma neutrophil counts



Article

Integration of biobank-scale genetics and plasma proteomics reveals evidence for causal processes in asthma risk and heterogeneity

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SUMMARY

Hundreds of genetic associations for asthma have been identified, yet translating these findings into mechanistic insights remains challenging. We leveraged plasma proteomics from the UK Biobank Pharma Proteomics Project (UKB-PPP) to identify biomarkers and effectors of asthma risk or heterogeneity using genetic causal inference approaches. We identified 609 proteins associated with asthma status (269 proteins after controlling for body mass index [BMI] and smoking). Analysis of genetically predicted protein levels identified 70 proteins with putative causal roles in asthma risk, including known drug targets and proteins without prior genetic evidence in asthma (e.g., GCHFR, TDRKH, and CLEC7A). The genetic architecture of causally associated proteins provided evidence for a Toll-like receptor (TLR)1-interleukin (IL)-27 asthma axis. Lastly, we identified evidence of causal relationships between proteins and heterogeneous aspects of asthma biology, including between TSPAN8 and neutrophil counts. These findings illustrate that integrating biobank-scale genetics and plasma proteomics can provide a framework to identify therapeutic targets and mechanisms underlying disease risk and heterogeneity.

INTRODUCTION

Asthma is one of the most prevalent chronic conditions and non-communicable diseases globally. The etiology of asthma is complex, with notable contributions from both genetic and environmental factors. Over 200 independent regions of the genome have been associated with asthma to date through genome-wide association studies (GWAS).^{1–3} While delineating the causal effector genes that mechanistically link genetic variation and disease risk could inform therapeutic development, doing so remains challenging due to the large proportion of associations in non-coding regions of the genome.^{4,5} Furthermore, asthma is a highly heterogeneous condition, with patients varying significantly in terms of clinical parameters such as age at onset, blood and airway eosinophil counts characteristic of T helper (Th)2 inflammation, and corticosteroid responsiveness.⁶ A genetic basis for this variation in asthma presentation has also been demonstrated.^{7–13} However, few effectively powered datasets with measurements of intermediate disease phenotypes are available, limiting the ability to make inferences on the mechanistic basis for this variation and how asthma risk genes may differentially influence disease presentation.

The wealth of biomedical and genetic data from the UK Biobank (UKB) has made it an invaluable resource for investigating the genetic architecture of a large array of binary and continuous traits, including asthma and features of asthma

presentation.^{1,7,8,14} The UKB resource has been bolstered by the measurement of 2,923 unique proteins from the plasma of 54,219 UKB participants by the UKB Pharma Proteomics Project (UKB-PPP).¹⁵ Plasma proteomic measurements reflect a wide range of biological and pathological processes, capture signals from a variety of cellular sources and tissues, and are strongly influenced by genetic variation.^{16,17} The data generated by the UKB-PPP, therefore, afford the ability to use plasma proteomic measurements as key molecular intermediates to link genetic variation to health outcomes and identify biomarkers or therapeutic targets.

Here, we leveraged UKB-PPP plasma proteomic data to investigate the asthma-associated plasma proteome and identify proteins causally related to asthma risk and/or heterogeneity. A summary of the key components of this study is provided in Figure 1. We observed significant dysregulation of the plasma proteome in individuals with asthma and applied a genetic causal inference framework with genetically predicted plasma protein levels to distinguish proteins for which variation could be causative versus reflective of disease. By examining the genetic architecture of proteins with evidence of causality, we gained insights into the biological mechanisms that may underlie these disease associations. Lastly, to explore if plasma proteomics could provide insights regarding asthma heterogeneity, we applied this framework to quantitative measures related to asthma heterogeneity, revealing differential associations



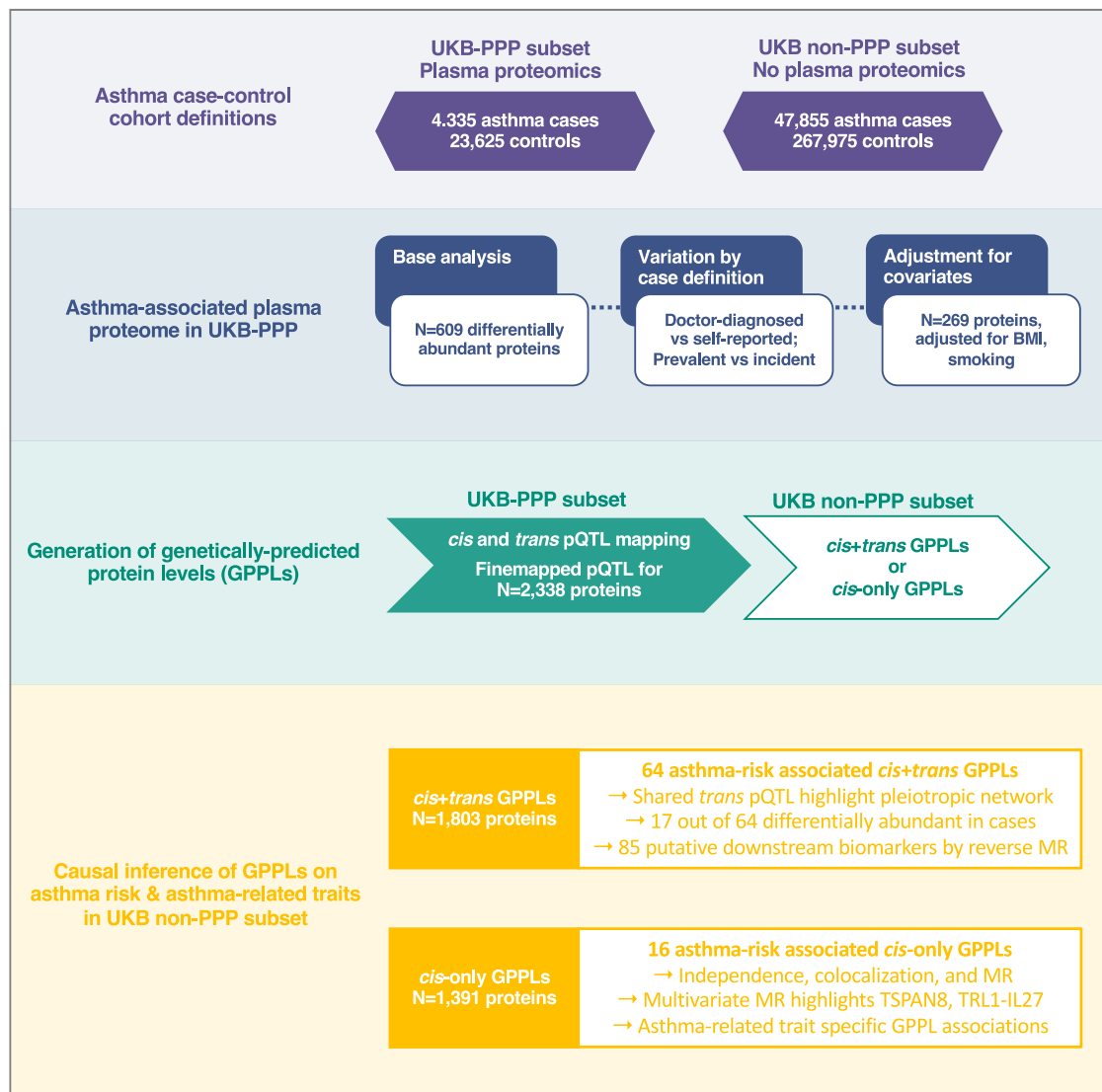


Figure 1. Study overview
Schematic outlining key components of this study.

between proteins and key aspects of disease presentation. These efforts collectively highlight how plasma proteomics and the genetic regulation thereof can be harnessed to uncover proteins and processes with putative causal relationships to asthma risk and heterogeneity.

RESULTS

Representation of asthma in the UKB-PPP

We identified asthma cases and non-asthma controls from the subset of 46,595 UKB participants randomly selected by the UKB-PPP for the generation of plasma proteomics using plasma taken at their baseline visit (Table 1, UKB-PPP subset; see STAR Methods). We also identified asthma cases and controls from the 400,115 UKB participants who were not randomly selected by

the UKB-PPP for the generation of plasma proteomics (UKB non-PPP subset) and could, therefore, serve as a non-overlapping subset for out-of-sample analyses. In the entire UKB, we identified 52,190 participants with doctor-diagnosed or self-reported asthma (diagnosis before or after their baseline visit) after excluding participants with evidence of chronic obstructive pulmonary disease (COPD) or idiopathic pulmonary fibrosis (IPF) (see STAR Methods). Of these participants with asthma, 4,335 were in the UKB-PPP subset and 47,855 were in the UKB non-PPP subset. We defined non-asthma controls as participants without a report of asthma, COPD, IPF, or atopy ($n = 23,625$ UKB-PPP controls and $n = 267,975$ UKB non-PPP controls). Based on Pan UKB-defined ancestry groups,¹⁸ 95% of UKB-PPP asthma cases were of European (EUR) ancestry ($n = 4,109$), with smaller numbers of individuals of African

Table 1. Asthma cases and controls in the UKB-PPP

	Asthma cases ^a (<i>n</i> = 4,335)	Controls (<i>n</i> = 23,625)	<i>p</i> ^b
Sex (% female)	58.9	53.1%	***
Age	56 (42–62)	59 (50–63)	***
Age of asthma onset (years) ^c	31 (12–47)	–	–
BMI	27.2 (24.4–30.5)	26.7 (24.2–29.8)	***
Eosinophils (cells/ μ L)	200 (100–300)	120 (100–200)	***
FEV ₁ % predicted ^c	91.9 (81.2–102.4)	98.0 (89.1–108.6)	***
Smoker (% at baseline visit)	7.8%	10.8%	***

The median and interquartile range are shown unless noted otherwise.

^aIncludes doctor-diagnosed and self-reported asthma cases.

^b $p \leq 0.05$ case versus controls with two-sided *t* test or Wilcoxon test.

^cData are only available for 58%–75% of participants with primary care records.

(AFR; *n* = 93), Central/South Asian (CSA; *n* = 84), Middle Eastern (MID; *n* = 18), East Asian (EAS; *n* = 20), and admixed American (AMR; *n* = 11) ancestry groups.

Asthma cases in the UKB-PPP subset were representative of the entirety of asthma cases in the UKB. Compared to controls, asthma cases in the UKB-PPP subset were more likely to be female, have a higher body mass index (BMI) and blood eosinophil counts, have reduced percentages of predicted forced expiratory volume (FEV₁% predicted), and were less likely to identify as smokers at the time of the baseline visit (Table 1). From the total of 4,335 UKB-PPP asthma cases, 2,694 participants had doctor-diagnosed disease, and the remainder were self-reported only. At the time of the baseline visit when the plasma used for proteomic measurement was sampled, 78% of UKB-PPP asthma cases had records of prevalent asthma (doctor diagnosed or self-reported). Of the 22% of UKB-PPP cases with reports of asthma after their baseline visit (incident asthma cases), 57% had reports of asthma within 10 years of their baseline visit.

Characterization of plasma proteins differentially abundant in asthma

Plasma proteins associated with disease status can include those for which variation is a consequence of disease status (i.e., downstream biomarkers), those related to various comorbidities and confounders, and others causally related to disease development.¹⁷ To define and characterize the plasma proteome associated with asthma status, we performed differential abundance analysis considering different sets of covariates and definitions of asthma status using the UKB-PPP sample set. We first performed linear regression within each of the Pan UKB-defined ancestry groups using individuals with prevalent or incident disease relative to their baseline visit and a base set of covariates (age, sex, batch, UKB assessment center, and the time between plasma collection and protein measurement). Among EUR ancestry asthma cases, we identified *n* = 609 proteins differentially abundant compared to controls ($p < 1.71 \times 10^{-5}$ Bonferroni corrected; Table S2). Top differentially abundant proteins included proposed asthma biomarkers (e.g., CHI3L1 and CCL26), markers of immune cells (e.g., CLC and CCL18), and proteins that are markers of asthma-related cell types in the lung (e.g., SCGB1A1 for airway epithelial cells).¹⁹ Pathway enrichment analysis on this set of differentially abundant proteins

highlighted disease-relevant processes related to inflammation and remodeling, including neutrophil degranulation and degradation of the extracellular matrix (Table S3). When considering only cases with doctor-diagnosed disease, we found the differential abundance effects were strongly correlated to those estimated when using a more liberal definition of either doctor-diagnosed or self-reported asthma cases ($r = 0.95$, $p < 2.2 \times 10^{-16}$; Figure S1A; Table S4), indicative of a conserved signature regardless of the source of case status.

Within non-EUR ancestry groups, no proteins were differentially abundant between cases and controls, likely due to a lack of power from much smaller sample sizes; additionally, none of the 609 proteins identified in the EUR ancestry analysis were significant in non-EUR ancestry groups at a lenient false discovery rate (FDR) < 0.2 threshold. To test for the general concordance of differential abundance between asthma cases and controls in EUR versus non-EUR ancestries, we computed the pairwise correlation of differential abundance effects between ancestries. We observed significant correlations between effects in EUR and AFR, CSA, or AMR ancestry groups (Pearson's $r = 0.69$, 0.66 , and 0.53 , respectively) but no correlation between EUR and ESA or MID ancestry groups (Pearson's $r = -0.2$ or -0.1 , respectively; $p > 0.05$). We note that two of the ancestries with the largest correlations (AFR and CSA) also have the largest sample sizes. Overall, this indicated patterns of shared changes in the asthma-associated plasma proteome across ancestries.

After establishing substantial differences in the plasma proteome with respect to asthma status, we next examined the impact of controlling for key anthropomorphic and behavioral traits confounded with asthma, such as BMI, smoking, and asthma-related medication use. We first controlled for BMI and smoking status to identify protein associations robust to two features commonly controlled for in molecular profiling studies of asthma due to strong confounding with asthma status. We identified *n* = 269 proteins differentially abundant when controlling for BMI and smoking status (Figure 2) in addition to the base set of covariates. Top differentially abundant proteins and pathways enriched among this set were similar to those found among the *n* = 609 proteins differentially abundant in the analysis with the base set of covariates (Table S5), indicating that asthma status was still the dominant signal. Adding BMI or smoking status to

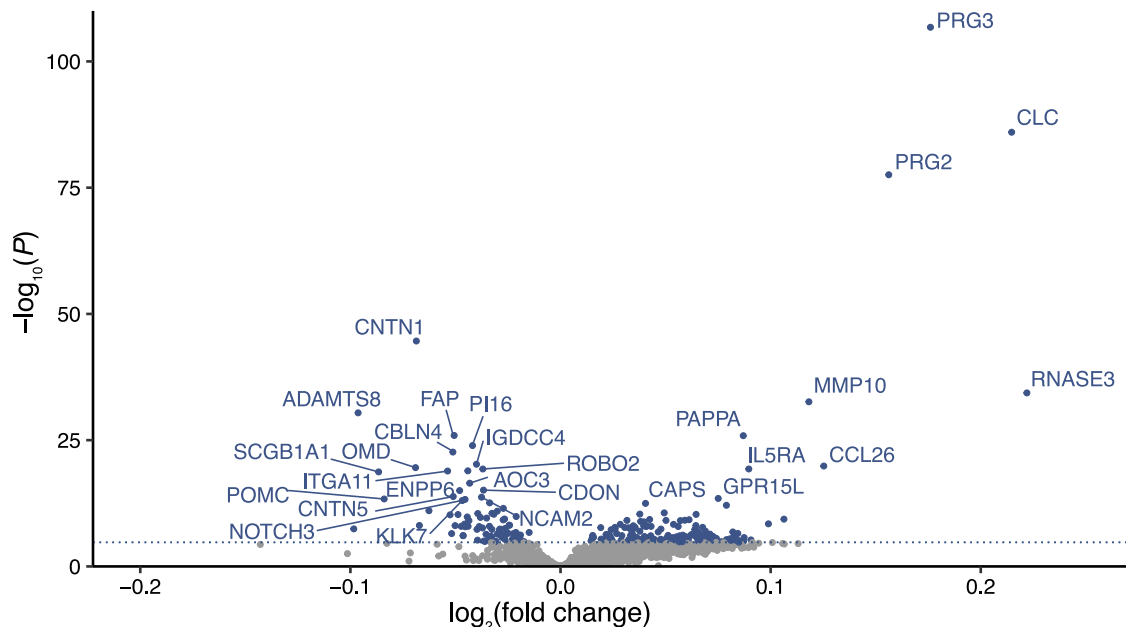


Figure 2. The UKB-PPP asthma-associated plasma proteome

Volcano plot of protein differential abundance between asthma cases ($n = 4,109$, European ancestry doctor diagnosed and self-reported) and controls ($n = 22,444$) controlling for base covariates (see main text), BMI, and smoking status. The horizontal line is at the Bonferroni-corrected $-\log_{10}(p)$. The top most significantly associated proteins are labeled.

the model individually also identified relationships with specific proteins of interest, such as the effects of BMI or smoking behavior on the differential abundance of MMP9, a proposed asthma biomarker (Figure S1B). Overall, the attenuation of protein differential abundance when controlling for BMI and smoking is consistent with the substantial effect of these variables on the plasma proteome regardless of the disease context and confounding with asthma status. However, it is also consistent with evidence of epidemiological relationships between BMI or smoking behavior and asthma risk and could represent the attenuation of true associations due to adjusting for variables that partially mediate the causal path to asthma.^{15,20,21} This complexity was further illustrated when controlling for asthma-related medication use, which resulted in an attenuation of the number of differentially abundant proteins from 609 to 469 proteins. This attenuation aligns with expectations that controlling for medication use by definition controls for case status. Among these 469 differentially abundant proteins, all but 1 protein (DXO) were significant in the medication-unadjusted analysis at $FDR < 0.2$ (97% significant at $FDR < 0.05$), indicating few new associations were identified when controlling for medication use.

Lastly, given that UKB-PPP asthma cases included both those with disease prevalent at the time of the plasma collection and incident to plasma collection, we sought to determine if the temporality of disease diagnosis could aid in identifying proteins for which differential abundance may be causal for, as opposed to a consequence of, asthma development. Overall, plasma protein fold changes between controls and cases with either prevalent or incident asthma were highly concordant ($r = 0.77$, $p < 2.2 \times 10^{-16}$; Figure S1C). This suggested that differential abundance in incident cases was not effective in distinguishing proteins

causative for disease versus those reflecting the disease state and that incident cases may have already been experiencing disease onset at the time of their plasma collection. In support of that inference, we found that 13% of cases defined as incident (either doctor diagnosed or self-reported) had a prescription for an inhaled asthma-associated medication at the time of their baseline visit, indicating that the records available to classify participants as having incident or prevalent disease are likely incomplete.

Collectively, these results highlight that the plasma proteome captures substantial differences between asthma cases and controls, including a number of disease-associated biological processes.

Generation of genetically predicted protein levels from *cis*- and *trans*-pQTLs

Given the limited ability of differential abundance to distinguish proteins with putative causal roles in asthma pathogenesis from those for which variation is a consequence of asthma, we adopted a causal inference framework based on the genetic regulation of the plasma proteome. Genetic effects on plasma protein levels reflect direct genetic regulation occurring in circulation or capture regulation in specific cellular or tissue contexts. Compared with many other influences on the plasma proteome, genetic effects enable causal inference because genetic variants are inherited independently of one another, (largely) fixed since conception, and not subject to environmental confounders.²²

To identify genetic instruments for plasma protein levels, we used the results of genome-wide protein quantitative trait locus (pQTL) mapping for 2,922 unique proteins performed by the UKB-PPP consortium.¹⁵ These analyses were restricted to the

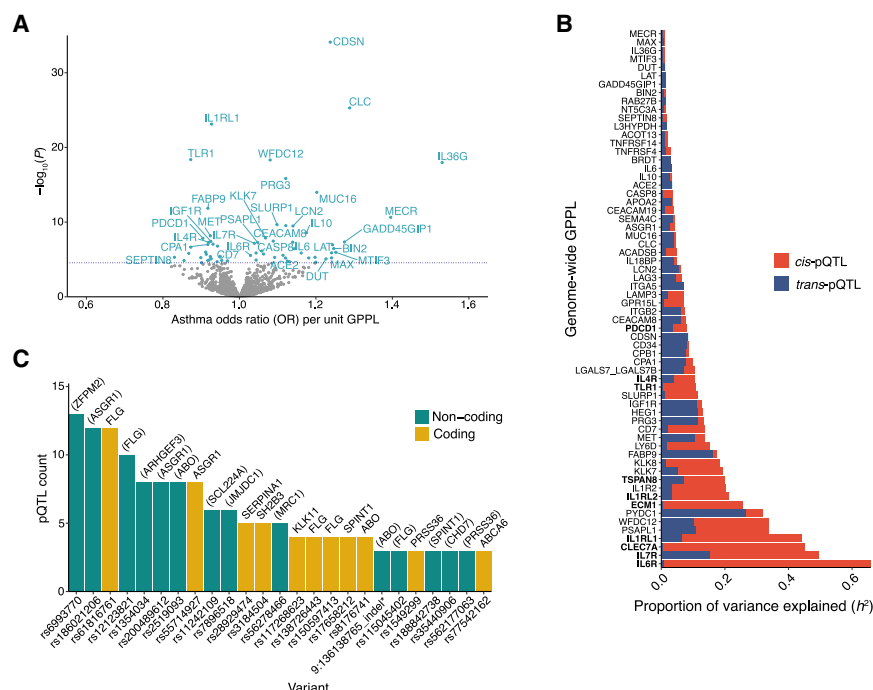


Figure 3. Causal inference with GPPLs identifies proteins associated with asthma risk

(A) Volcano plot of association between *cis+trans* GPPLs and asthma risk in $n = 45,808$ asthma cases and $n = 254,107$ controls in UKB non-PPP subset. The horizontal line is at the Bonferroni-corrected $-\log_{10}(P)$.

(B) Proportion of variation in measured protein levels explained by GPPL *cis*-pQTLs (red) or *trans*-pQTLs (navy). Proteins with *cis*-GPPLs associated with asthma are in bold.

(C) Number of *trans*-pQTLs from asthma-associated *cis+trans* GPPLs that map to the same variant. Only variants with 3 or more *trans*-pQTLs are shown. Coding variants (yellow) and non-coding variants (green).

protein levels to avoid weak instrument bias.²³ This set of *cis+trans* GPPLs was derived from a median of 12 pQTL instruments and explained a median of 7.4% of variation in protein levels. In total, we identified 64 *cis+trans* GPPLs significantly associated with asthma risk (Figure 3A; $p < 2.8 \times 10^{-5}$ Bonferroni corrected; Table S6).

GPPL effects on asthma risk using both doctor-diagnosed and self-reported cases were strongly correlated to those when association testing was performed on only cases with doctor-diagnosed disease ($n = 31,506$ cases; $r = 0.99$, Figure S2). The median percentage of variation in measured protein levels explained by asthma-risk-associated *cis+trans* GPPLs was 7.0% (range: 1%–66%; Table S7), with differing proportions of variance explained in *cis* or *trans* (Figure 3B). Due to the limited ability to generate ancestry-specific GPPLs from the small numbers of UKB-PPP participants of non-EUR ancestries and poor transferability of GPPLs across ancestral groups,^{2,24} we were precluded from informative multi-ancestry GPPL analyses. However, we did note that 25 of the proteins identified in the GPPL analysis have one or more pQTLs reported in both EUR ancestry and non-EUR ancestry UKB-PPP cohorts.¹⁵ Furthermore, among the 64 with asthma-associated *cis+trans* GPPLs, 8 proteins, including CD7, CLEC7A, and interleukin-18 binding protein (IL-18BP), had at least one pQTL only identified in UKB-PPP non-EUR ancestry populations, raising the possibility of ancestry-enriched regulatory mechanisms for these proteins.¹⁵

The proteins represented by these 64 asthma-risk-associated *cis+trans* GPPLs included several directly related to targets in clinical trials for asthma (i.e., IL-1RL1/ST2, IL-4R, and IL7R), with effects directionally consistent with therapeutic hypotheses.^{25–28} These findings highlighted the utility of this approach to identify therapeutically informative associations. Among the other significantly associated *cis+trans* GPPLs, several represent proteins with functional evidence from *in vivo* and/or *in vitro* models of asthma, including PDCD1/PD-1, CLEC7A/Dectin-1, CD34, and SEMA4C.^{29–34} Additionally, asthma-risk-associated *cis+trans* GPPLs included proteins linked to a diverse

subset of the UKB-PPP who were of EUR ancestry and selected at random from the UKB ($n = 34,557$) rather than on the basis of phenotype (as previously described¹⁵). We identified fine-mapped pQTL credible sets for 2,338 proteins, with a median of 4 independent signals in *cis* and 5 in *trans* per protein. For each protein, we then used the most probable set of putatively causal independent variants to generate genetically predicted protein levels (GPPLs) for each protein across UKB participants (see STAR Methods). GPPLs represent the heritable component of variation in protein levels as estimated by the aggregate effect of pQTLs. Therefore, GPPLs can be used as instruments to test for associations between unconfounded, genetically predicted levels of plasma proteins and disease outcomes in independent subsets of individuals, such as those in the UKB non-PPP subset. GPPLs derived from both *cis*- and *trans*-pQTLs (“*cis+trans* GPPLs”) explained a median of 4.5% of the variation of protein levels in the UKB-PPP subset (where we had both measured protein levels and derived GPPLs). The explanatory power ranged greatly across proteins (0.1%–80%). Among all *cis+trans* GPPLs, a greater proportion of estimated variant heritability was explained by *cis*-pQTLs compared to *trans*-pQTLs (61% versus 39%, respectively).

Causal inference with GPPLs identifies proteins associated with asthma risk

To make inferences regarding plasma proteins that denote causal processes in asthma, we used logistic regression to model asthma status as a function of individual *cis+trans* GPPLs in UKB non-PPP asthma case and control participants of EUR ancestry ($n = 45,808$ doctor-diagnosed or self-reported asthma cases, $n = 254,107$ controls). We focused on the 1,803 *cis+trans* GPPLs that explained at least 1% of the variation in

set of asthma-related processes, including ECM1 related to Th2 signaling,^{35,36} Toll-like receptor (TLR)1 related to innate and adaptive immunity,³⁷ RAB27B related to mast cell degranulation,^{38,39} and TSPAN8 related to airway mucin secretion.⁴⁰

We explored if there was evidence that these GPPLs represent distinct associations with asthma risk or captured broader programs of genetic regulation through pQTL networks. In total, there were 876 unique fine-mapped pQTL variants across these 64 asthma-risk-associated *cis+trans* GPPLs, with 77 variants associated with more than one protein. We hypothesized that shared *trans*-pQTLs may represent pleiotropic effects driving broader biological processes. We identified 25 pQTLs associated with the levels of 3 or more of the 64 proteins (Figure 3C) and 24 shared pQTLs for which the putatively causal variant was a coding variant, pointing toward likely causal effectors of the *trans* effects (Table S8). The genes harboring these coding variants included established asthma risk genes such as *FLG*, with the *trans*-pQTL at *FLG* coding variants colocalizing with an association for asthma risk (H4 posterior probability [PP] > 0.85). *FLG* encodes filaggrin, a key protein in epidermal barrier protection that is implicated in allergy and atopic dermatitis in addition to asthma.⁴¹ Other coding variants included those in *CFTR* and *SERPINA1*, two genes associated with other pulmonary diseases and more severe forms of asthma,^{42,43} as well as in the serine proteases *PRSS36*, *SPINT1*, and *KLK11*. While important roles for serine proteases in asthma have been reported, we have not found previous genetic evidence linking these genes to asthma.⁴⁴

Of the 64 proteins represented by these asthma-risk-associated *cis+trans* GPPLs, only 17 were differentially abundant between controls and cases in our cross-sectional analysis with or without correction for BMI and smoking. This finding highlighted that differential abundance has poor sensitivity and specificity for the detection of proteins with putatively causal roles. We tested if the asthma-GPPL associations may have resulted from reverse causation (i.e., genetic effects underlying asthma risk influence the levels of these proteins) by performing two-sample Mendelian randomization (MR) with asthma as the exposure and protein levels as outcomes. We selected instrumental variables (IVs) for asthma from an asthma GWAS generated from the UKB non-PPP subset of participants to maintain independence from the UKB subset used for pQTL mapping (see STAR Methods; Table S9). Using the inverse-variance-weighted (IVW) estimate and weighted median methods, as well as MR-Egger⁴⁵ and MR-PRESSO⁴⁶ sensitivity tests for pleiotropy and outliers, we identified significant associations between asthma risk and the levels of 3 of the 64 proteins with asthma-risk-associated *cis+trans* GPPLs (CEACAM8, CLC, and PRG3). These three proteins were also among those differentially abundant between cases and controls, a pattern consistent with effects from reverse causation. Overall, these findings indicated that the majority of the associations between *cis+trans* GPPLs and asthma risk were not driven by reverse causation.

After establishing that the associations between asthma risk and *cis+trans* GPPLs were, in most cases, not driven by reverse causation, we also leveraged this two-sample reverse MR framework (i.e., asthma as the exposure and protein levels as

the outcome) to all measured proteins to identify putative downstream biomarkers of asthma. In total, we identified 85 proteins with significant associations by MR (Table S10). As expected of proteins for which differential abundance is a consequence of asthma status, all of these proteins were differentially abundant in asthma. This set included proposed biomarkers for asthma, such as CHI3L1/YLK-40, RNASE3/ECP, CCL26, and MMP9⁴⁷ (Figure S3). Additionally, several proteins in this set, including CLC, LEP, and CA14, were top features in models generated to predict incident asthma from plasma proteomics and clinical records.^{48,49} This indicated that these putatively downstream biomarkers may have potential as predictive biomarkers as well. While a causally upstream versus downstream role of CHI3L1 in asthma and genetic evidence thereof has been debated in the literature,^{50–52} our MR and GPPL analyses only found support for CHI3L1 as a downstream biomarker and indicated that the aggregate pQTL component of CHI3L1 levels (as detectable in plasma) is not associated with asthma risk. In total, this analysis not only confirmed that variation in the levels of proposed plasma biomarkers is secondary to the development of asthma but also yielded an expanded set of putative plasma downstream biomarkers of asthma for which there is genetic support.

Collectively, these findings demonstrate the ability to use genetic regulation to identify proteins with evidence of being putatively causal in asthma risk and distinguish them from proteins for which variation is a consequence of disease.

Local genetic regulation aids in interpretation of asthma-risk-associated GPPLs

While we first focused on GPPLs derived from both *cis+trans* pQTLs given their strong explanatory power, *trans*-pQTLs are also prone to indirect pleiotropic effects that can compromise interpretability. Therefore, we also performed asthma risk association testing with GPPLs derived only from *cis*-pQTLs (*cis*-GPPLs), which explained at least 1% of the variation in protein levels ($n = 1,391$). These 1,391 *cis*-GPPLs were derived from a median of 6 *cis*-pQTL instruments, explained a median of 6.3% of the variation in protein levels, and captured 82% of estimated causal variant heritability.

We identified 16 *cis*-GPPLs significantly associated with asthma risk ($p < 3.6 \times 10^{-5}$ Bonferroni corrected; Figure 4A; Table S11; median: 9.5% of the variation explained [range: 1.1%–65.2%]). The effect estimates from these significantly associated *cis*-GPPLs were strongly correlated with the effect estimates from their *cis+trans* GPPLs due to a large proportion of GPPL explanatory power coming from *cis* regions for these proteins ($r = 0.96$; $p = 3.9 \times 10^{-9}$; Figure 4B). Of the 16 proteins with asthma-risk-associated *cis*-GPPLs, 6 proteins did not reach significance in the *cis+trans* GPPL analysis due to divergent effects of the *trans*-GPPL component and *cis*-GPPL component on asthma risk (Figure S4). This brought the total number of unique proteins with asthma-risk-associated GPPLs to 70 (54 only associated with *cis+trans* GPPLs, 10 with both *cis+trans* and *cis*-only GPPLs, and 6 only associated with *cis*-GPPLs). For these 6 new proteins with *cis*-GPPL associations, we again tested for reverse causation using two-sample MR (asthma risk as the exposure and protein levels as the outcome) and

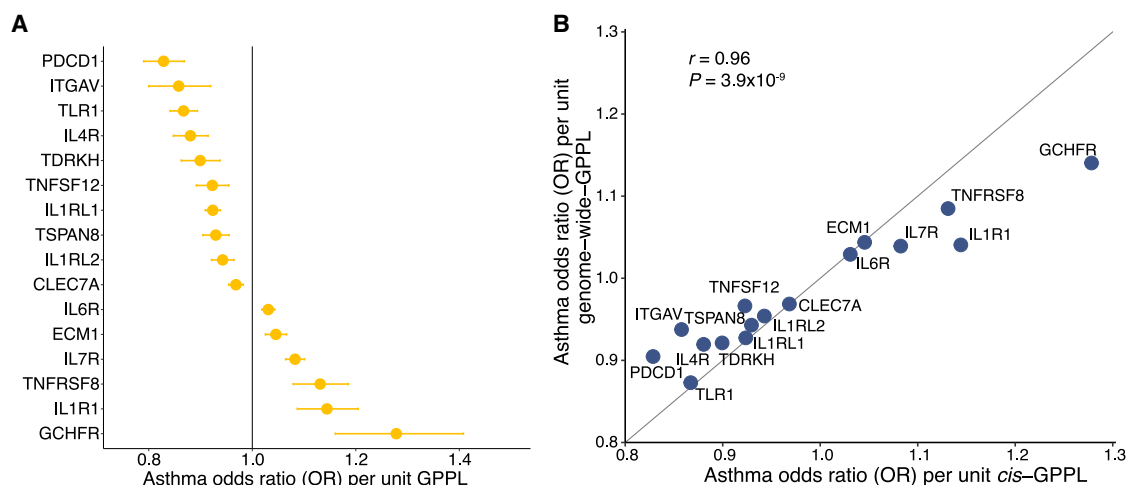


Figure 4. Identification of 16 *cis*-GPPLs associated with asthma risk

(A) Forest plot of 16 *cis*-GPPLs associated with asthma risk ($p \leq 3.58 \times 10^{-5}$) in $n = 45,808$ asthma cases and $n = 254,107$ controls in UKB non-PPP subset. Beta and 95% confidence intervals are depicted.

(B) Correlation between effects of *cis*-GPPLs (x axis) and *cis+trans* GPPLs (y axis) on asthma risk.

only observed a significant association for TNFSFR8, indicating that TNFSFR8 levels have a more complex relationship with asthma risk (Figure S5). While many of these asthma-risk-associated *cis*-GPPLs represent genes that are predicted effector genes at asthma GWAS loci, to our knowledge, *CLEC7A*, *GCHFR*, and *TDRKH* have not previously been implicated in asthma risk through a genetic approach.

We next sought to explore the nature of these *cis*-GPPLs and asthma risk associations by assessing their independence, the consistency of composite variant effects, and colocalization between pQTL and asthma risk. First, although limiting GPPLs to *cis*-pQTLs only reduces the influences of pleiotropy, we noted that 3 of the 16 asthma-risk-associated *cis*-GPPLs, IL-1RL1, IL-1RL2, and IL-1R1, are located sequentially in the genome within 250 kb of each other. This raised the possibility that one or more of these associations may be confounded by shared local genetic effects. To test for independence among the 16 *cis*-GPPLs, we identified instances in which these proteins shared a fine-mapped *cis*-pQTL and ran multivariate asthma association analyses where the *cis*-GPPLs of other proteins were included in the model. The associations for IL-1RL1, IL-1RL2, and IL-1R1 were not attenuated in the multivariate GPPL analysis, indicating they represent independent signals at this locus (Figure S6). Furthermore, multivariate analysis at this locus revealed a significant association between asthma risk and IL-18R1 *cis*-GPPLs, which was not significant in the univariate model ($p = 3.4 \times 10^{-12}$ multivariate versus 4.1×10^{-2} univariate). We determined the IL-18R1 *cis*-GPPL association with asthma risk was dependent upon a correction for IL-1RL1 *cis*-GPPL effects. These findings are consistent with the complex genetic relationship for atopic dermatitis risk at this locus.¹⁶ We did not find any other instances in which the *cis*-GPPL associations were modified in multivariate analyses, indicating that the 16 asthma-risk-associated *cis*-GPPLs represented independent signals.

We next performed colocalization analysis between summary statistics for the 16 proteins with asthma-risk-associated *cis*-GPPLs and asthma. Evidence of colocalization where two traits share at least one putatively causal variant (H4 PP > 0.8) can provide additional support for a causal relationship between two traits. Alternatively, colocalization can provide evidence that associations are driven by distinct genetic variants in linkage disequilibrium (LD; i.e., plasma protein levels and asthma risk have distinct causal variants in LD; H3 PP > 0.8). Using coloc-SuSiE,^{53–55} which allows for the existence of multiple causal variants at a locus, we identified at least one pQTL colocalizing with asthma risk for 7 of the 16 proteins (H4 PP > 0.80). Of proteins with supportive evidence from MR, TSPAN8, TLR1, and IL-6R all showed evidence of H4 colocalization. Additionally, we found evidence for H4 colocalization for *CLEC7A* despite asthma GWAS variants near *CLEC7A* being below genome-wide significance in both our GWAS and larger, previously reported asthma GWAS.^{1,56} For all other asthma-risk-associated *cis*-GPPLs, we found evidence that both protein levels and asthma had genetic association signals in the same regions but did not necessarily share the same putatively causal variant (H3 PP > 0.80).

Lastly, while allelic scores such as GPPLs can increase power by representing aggregate genetic effects,⁵⁷ they do not describe the consistency or specificity of variant effects. Therefore, we also performed two-sample MR to further examine the evidence of a causal relationship between protein levels (exposure) of the 16 proteins with asthma-risk-associated *cis*-GPPLs and asthma risk (outcome). We used the independent pQTLs that compose the *cis*-GPPLs as IVs after assessing their satisfaction of the MR assumptions of relevance, exclusivity, and exclusion²² (see STAR Methods). The IV effects on asthma risk were estimated from the asthma GWAS in the UKB non-PPP subset of participants. Using the IVW estimate MR method, we identified significant effects for 10 proteins, yet MR-Egger⁴⁵ and MR-PRESSO⁴⁶ sensitivity analyses designed to test for

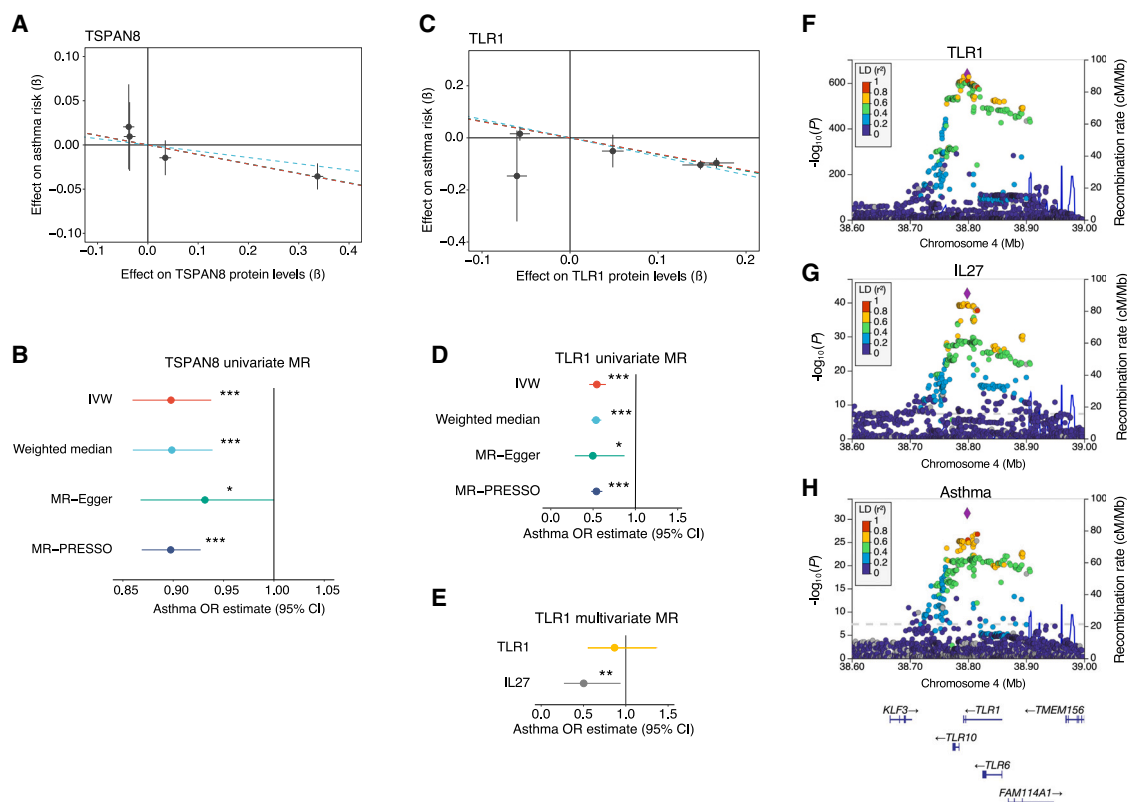


Figure 5. Mendelian randomization highlights genetic effects on TSPAN8 levels and a TLR1-IL27 axis in asthma risk

(A and B) Effects of *cis*-pQTLs on protein levels (x axis) and asthma risk (y axis), with dashed lines representing point estimates of univariate MR methods for (A) TSPAN8 and (B) TLR1. Estimates for IVW (red), weighted median (blue), MR-Egger (green), and MR-PRESSO (navy) are shown.

(C and D) Point estimates from univariate MR methods for TSPAN8 (C) and TLR1 (D). *** $p \leq 0.005$ and * $p \leq 0.05$.

(E) Point estimates of effects from TLR1 multivariate MR, including IL-27. ** $p \leq 0.05$.

(F–H) Locus zoom of genetic associations for TLR1 and IL-27 protein levels and asthma risk at TLR1 locus.

pleiotropy revealed the heterogeneity of IV effects for 7 of these 10 proteins (Table S12). Three proteins, TSPAN8, TLR1, and IL-6R, demonstrated significant associations across all MR methods, were robust to additional tests for sensitivity, and were among those with evidence of H4 colocalization to asthma. For TSPAN8, we found evidence that lower TSPAN8 levels are associated with asthma risk (Figures 5A and 5B; Table S13). While a specific role of TSPAN8 in asthma development has not been reported, it has been shown to be a negative regulator of airway mucin secretion *in vitro*,⁴⁰ which is directionally consistent with increased airway mucus production as a pathological component of asthma. Our significant associations for TLR1 and IL-6R replicated previously reported associations for reduced TLR1 levels (Figures 5C and 5D) and increased IL-6R levels (Figure S7) on asthma risk.^{58,59} Collectively, these findings provided further evidence for causal roles of specific proteins with consistent plasma pQTL effects.

In total, our *cis*-GPPL analysis identified proteins with evidence of causal roles in asthma driven by *cis*-acting regulation alone, including several proteins not identified in the *cis*+*trans* analysis. Additional interrogation of the *cis*-GPPL associations through colocalization and MR also provided further causal support for some of these proteins. However, we also note that pro-

teins with extensive prior functional evidence, such as IL-1RL1, CLEC7A, and PDCD1, exhibited H3 colocalization and/or heterogeneity in MR. This raised the possibility that for certain proteins, we may have limited inference ability with colocalization and MR due the sensitive task of selecting IVs, the probabilistic nature of fine-mapping and colocalization, and tissue- or context-specific variant effects.^{60,61} For example, it remains possible that proteins with H3 colocalization are indeed causally related to asthma risk, yet the regulatory or cellular contexts in which these proteins are causal are not faithfully captured in plasma (see discussion).

Genetic architecture of GPPLs reveals evidence for direct and mediated effects on asthma risk

To gain a greater mechanistic understanding on how proteins with asthma-risk-associated *cis*-GPPLs influence disease development, we sought evidence of vertical pleiotropy involving other measured proteins, as this could aid in the identification of mediators of variant effects. Specifically, we explored if proteins with asthma-risk-associated *cis*-GPPLs had a *cis*-pQTL associated in *cis* or *trans* with the levels of another protein measured in the UKB-PPP. Three of the 16 proteins with asthma-risk-associated *cis*-GPPLs, TLR1, IL-7R, and TNFSF12, met this criterion.

TLR1 and IL-7R both had a *cis*-pQTL that colocalized with a *trans*-pQTL for another protein (IL-27 and CD48, respectively). TNFSF12 had a *cis*-pQTL at rs186021206, a known pQTL “hot-spot” associated with the levels of over 200 proteins, including ASGR1 encoded in *cis*.¹⁵ Given that TLR1 showed evidence of a direct effect on asthma risk through our MR analysis, the shared association with IL-27 levels in *trans* raised the possibility that IL-27 may mediate or modify the association between genetic effects on TLR1 and asthma risk. To test this hypothesis, we performed two-sample multivariate MR, with the effects of IVs on the secondary-exposure IL-27 identified in *trans* (i.e., mediator) included in the model.⁶² The significant main effect of TLR1 levels was attenuated when including IV effects on IL-27, indicating that IL-27 levels likely mediate the putative relationship between TLR1 and asthma risk (Figure 5E). Furthermore, we identified multi-trait colocalization between TLR1, IL-27, and asthma risk at the TLR1 locus (Figures 5F-H), supporting a shared relationship between these proteins and asthma risk.

We also sought to determine if *trans*-pQTL signals could provide additional regulatory or biological insights into the 16 proteins with asthma-risk-associated *cis*-GPPLs. We explored this through two different approaches. We first asked if a pQTL acting in *trans* for one of the 16 proteins was also associated in *cis* with another protein, positing that the protein associated in *cis* may represent an upstream regulator. This revealed a *trans*-pQTL for CLEC7A that colocalizes with a *cis*-pQTL for CSF3/G-CSF, indicating that CSF3, an important contributor to neutrophilic inflammation in asthma,⁶³ may regulate CLEC7A levels. As a second approach, we tested if proteins sharing at least one *trans*-pQTL with any of the 16 asthma-risk-associated *cis*-GPPLs were enriched for biological functions or pathways. An example of this was observed for a set of 56 proteins sharing a *trans*-pQTL with TSPAN8. Proteins in this set include those with well-described roles in cell-cell adhesion or barrier function, such as EPCAM, CDH1, and CXADR, with which measured plasma TSPAN8 levels are also significantly correlated (Table S14). Furthermore, pathway enrichment analysis of this protein set highlighted functions related to cell surface interactions and leukocyte migration, indicating that TSPAN8 may have a role in the lungs beyond regulation of mucus secretion (Table S15). Such a role for TSPAN8 is consistent with a report of TSPAN8 playing a role in inflammation and integrity of the intestinal epithelium.⁶⁴

In total, the results of this analysis illustrate how the genetic regulation of proteins can shed light on potential mechanisms by which these proteins signal to influence asthma pathogenesis.

Genetic effects on plasma protein levels highlight distinct associations with features of asthma

In addition to the abundant number of GWAS associations supporting a genetic basis of asthma risk, a genetic basis for variation in disease presentation across patients has also been reported.^{7–10} Variation in asthma presentation includes significant clinical heterogeneity in terms of quantitative characteristics (i.e., blood cell counts, FEV₁% predicted, and age of onset) as well as therapeutic response,⁶⁵ consistent with a model where

multiple pathophysiological processes are involved in asthma presentation, each of which maps to partially distinct genetic and/or environmental etiologies. To gain further insights into the mechanistic basis of asthma heterogeneity, we therefore extended our analysis to identify proteins with evidence of causal effects on quantitative features of asthma presentation. We first assessed if the 16 *cis*-GPPLs associated with asthma risk were also associated with features of asthma by testing the association between those *cis*-GPPLs and reported asthma age of onset; counts of circulating blood eosinophils, neutrophils, or lymphocytes; and FEV₁% predicted. We performed this analysis using linear modeling of baseline visit data from UKB non-PPP asthma cases. Four *cis*-GPPLs, TLR1, TDRKH, IL-1RL2, and IL-1R1, were significantly associated with asthma age of onset ($p < 3.5 \times 10^{-4}$; Figure 6). TLR1 *cis*-GPPLs had the strongest association, such that lower genetically predicted levels of TLR1 were associated with a younger age of asthma onset, consistent with a role for TLRs in early-life immune sensitization.⁶⁶ Six *cis*-GPPLs were associated with eosinophil count, including those with known roles in eosinophilic inflammation, such as IL-4R, but also those with less-well-recognized roles in asthma, such as GCHFR. TSPAN8 and TLR1 *cis*-GPPLs were the only associations for counts of neutrophils and lymphocytes, respectively. No asthma-risk-associated *cis*-GPPLs were associated with FEV₁% predicted.

We were also interested in determining if the *cis*-GPPLs associations with blood cell counts were specific to individuals with asthma or more broadly represented genetic effects on these traits in the general population. In UKB non-PPP controls, we found many of the same associations between *cis*-GPPLs and blood cell count traits as observed in individuals with asthma, representing shared mechanisms regardless of disease status (Figure 6). Two exceptions were associations for GCHFR with eosinophils and TSPAN8 with neutrophils, which were restricted to individuals with asthma. The association between TSPAN8 *cis*-GPPLs and neutrophils exclusive to individuals with asthma led us to consider if this relationship could be explained by confounding variables between cases and controls, such as BMI, which is also associated with blood neutrophil counts. However, we did not find evidence that *cis*-GPPLs for TSPAN8 were associated with BMI in asthma cases or controls ($p = 0.86$ and $p = 0.27$, respectively), indicating that differences in BMI were not driving this association. Overall, these findings highlighted that many, but not all, genetic associations for asthma-related quantitative traits are shared between cases and controls.

Lastly, as genetic factors beyond those influencing disease risk can also influence disease presentation, we also performed association testing between quantitative features of asthma and the full set of *cis*-GPPLs (including those not significantly associated with asthma risk). We identified additional *cis*-GPPLs associated with blood cell traits in both asthma cases and controls (Figure S8; Table S16), as well as five *cis*-GPPLs significantly associated with FEV₁% predicted among controls. Among the associations with FEV₁% predicted, all except SIPRA align with known GWAS associations for lung function or COPD.^{56,67} These associations represented a broader set of putative genetic regulators of asthma heterogeneity that are independent from regulators of disease risk.

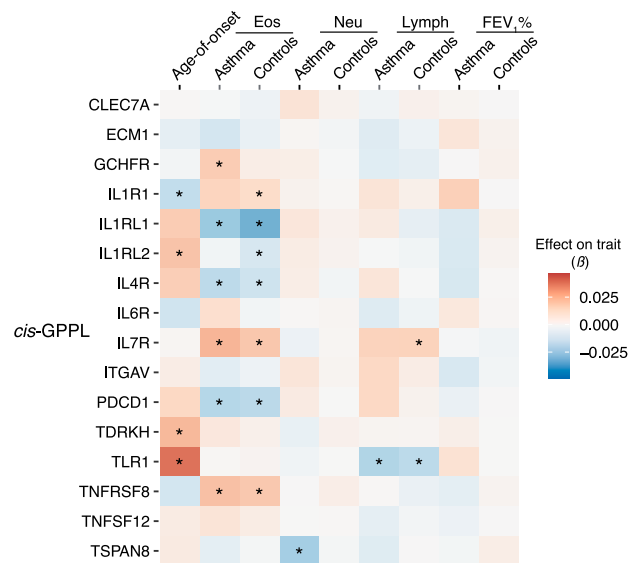


Figure 6. Asthma-risk-associated GPPLs are differentially associated with quantitative traits heterogeneous in asthma

Heatmap depicting the effect of 16 asthma-associated *cis*-GPPLs on asthma age of onset or quantitative traits in either $n = 45,808$ asthma cases or $n = 254,107$ controls in UKB non-PPP subset (Eos, eosinophil count; Neu, neutrophil count; Lymph, lymphocyte count). * $p < 3.5 \times 10^{-4}$ (Bonferroni corrected).

Collectively, these results highlight the ability to associate proteomic factors to key features of asthma heterogeneity and that proteins with genetic evidence for causal roles in asthma risk vary in terms of their connection to disease presentation.

DISCUSSION

Integrating plasma proteomic data with the rich phenotypic and genetic profiling of the UKB enabled us to identify putative plasma proteomic biomarkers of asthma and proteins with evidence of causal roles in asthma. A prior meta-analysis on plasma proteomic measurements of 4,860 proteins from 1,231 adults with asthma identified 115 differentially abundant proteins. Of these 115 proteins, 67 were also measured by the UKB-PPP, and 36 were differentially abundant in our analysis when controlling for BMI and smoking status (43 differentially abundant in our study when using base covariates).⁶⁸ While, in total, we identified hundreds of proteins differentially abundant between asthma cases and controls, causal inference with GPPLs yielded only a select subset of plasma proteins with evidence of causal roles in disease risk. This finding was consistent with the notion that differences in gene products, at least as measured in plasma, more often reflect processes secondary to disease pathology or confounders rather than those causal for disease.⁶⁹

The subset of proteins with causal evidence included proteins related to asthma drug targets as well as those involved in a variety of biological mechanisms and pulmonary cell types linked to asthma, including inflammation, extracellular matrix integrity, and mucus secretion. Given that plasma protein levels are influenced by the cellular activity of tissues across the body, whether

these proteins with causal evidence exert their effects on asthma risk in plasma or through cellular processes in other tissues (i.e., the lung) remains to be determined. Many of these protein associations with asthma risk would have been missed if we restricted our analysis to proteins exhibiting differential abundance between cases and controls, highlighting the importance of taking an unbiased approach. The *cis*-GPPL associations we identified for IL-1R1, IL-7R, ECM1, IL-6R, TNFSF12, IL-1RL1, and TLR1 replicate previously reported findings for asthma in independent populations with genetic-proteomic approaches.^{70,71} Additionally, the expanded set of protein measurements and resulting pQTLs from the UKB-PPP dataset allowed us to identify causal evidence for TSPAN8 through MR, which, to our knowledge, has not been previously tested in an MR framework for asthma.

Leveraging pleiotropic effects on plasma protein levels in both *cis* and *trans* allowed us to generate new mechanistic hypotheses linking these proteins to asthma risk or heterogeneity. For example, using multivariate MR, we found evidence for a genetically driven TLR1-IL-27 axis in asthma risk. While the TLR1 locus is a replicated GWAS locus for asthma, far less is known about the role of TLR1 in asthma compared to other proteins in the TLR family, such as TLR2, TLR3, and TLR4.⁶⁶ Of these additional TLRs, we only identified *cis*-pQTLs for TLR3 and TLR4 explaining 79% and 2% of the variance in protein levels, respectively, yet their *cis*-GPPLs were not significantly associated with asthma risk. Consistent with our findings, TLR1 has been shown to induce IL-27, a heterodimeric cytokine composed of IL-27A/p28 and EBI3.⁷² Patients with untreated asthma have been reported to have lower levels of plasma IL-27, and IL-27 has also been shown to suppress inflammation and other endpoints in animal models of asthma.^{73–75} The negative association between IL-27 levels and features of disease is consistent with the negative relationship we found between the TLR1 or IL-27 pQTL and asthma risk.

As a second example of biological findings resulting from pleiotropic effects on the plasma proteome, we found evidence that TSPAN8 may have a role in barrier integrity and leukocyte migration through the pathway enrichment of proteins with shared genetic regulation in *trans*. Such a role could explain the association we observed between TSPAN8 GPPLs and increased neutrophil count in individuals with asthma. Whether the effect of TSPAN8 on blood neutrophil count in individuals with asthma is related to the demonstrated role of TSPAN8 in mucus secretion remains unclear,⁴⁰ but such a relationship could result from direct interactions between airway mucus and neutrophils or reflect an indirect effect from glucocorticoid effects on blood neutrophils.^{76,77} In this way, it is possible that the relationship between TSPAN8 GPPLs and blood neutrophils in asthma points toward either a modifier of the disease state or a subset of patients with a specific pathology.

Using allelic scores rather than individual variants as instruments in our initial approach to discover asthma-risk-associated proteins enabled us to identify proteins for which the collective aggregate effects of variants are associated with asthma risk, yet we also observed heterogeneity among individual variant effects. Heterogeneity among variant effects is often suspected to arise from either horizontal or vertical pleiotropy, which violates

MR assumptions of IVs.²² Since plasma proteomics captures genetic effects that may reflect the activity of multiple tissues, heterogeneity in variant effects may represent tissue-specific pQTL effects, including those in tissues not causally linked to the outcome of interest. While in comparison to *trans*-pQTLs, *cis*-pQTLs have been shown to be more likely to be shared across tissues, the effects of tissue-specific *cis*-pQTLs or *cis*-eQTLs can be observed in plasma protein levels.^{78,79} One example of this has been reported for IL-1RL1 in the context of asthma, where IL-1RL1 plasma protein *cis*-pQTLs were shown to align with *cis*-eQTLs in specific lung compartments but not in *cis*-eQTLs from whole blood.⁸⁰ This additionally highlighted that plasma pQTLs can capture genetic effects originating in specific disease-relevant cell types, such as those in the lung.⁸⁰ Future efforts to generate tissue- or cell-type-specific pQTL atlases (including those across contexts) will be valuable toward disentangling heterogeneity in variant effects captured in plasma or whole tissues. At the same time, it cannot be assumed that all disease-relevant pQTL are detectable in plasma, which is a limitation when performing colocalization between plasma pQTLs and asthma risk loci.

Additionally, we were able to examine genetic and proteomic effects not only on asthma risk but also on features variable across individuals with asthma that are associated with clinical outcomes. We found that a majority of asthma-risk-associated *cis*-GPPLs were also associated with increased blood eosinophils in both cases and controls, consistent with findings that asthma heritability is enriched for signatures of Th2 inflammation and previously reported evidence from MR that blood eosinophil count may be causally related to asthma development.^{81,82} However, asthma risk-associated *cis*-GPPLs were not universally associated with blood eosinophil or lymphocyte count, demonstrating the ability to identify proteins that may influence asthma risk through other mechanisms. As patients with asthma with low eosinophil count and disease classified as “Th2-low” or “non-Th2” may have more severe disease and are less likely to respond to biologics targeting the Th2 pathway, genetic insights into the non-Th2 mechanisms of asthma remain important for therapeutic development. We further found evidence for both Th2-related and Th2-unrelated mechanisms when examining shared *trans*-pQTL effects among asthma-risk-associated proteins. For example, proteins with pQTLs colocalizing to the pleiotropic *FLG* locus may be related to the atopic march (a typically Th2-related process),^{41,83} while proteins with pQTLs driven by putatively causal coding variants in *SERPINA1* and *CFTR* may be reflective of processes driven by the activity in a broader range of cell types, such as alveolar or epithelial cells.⁸⁴

Limitations of the study

There are important limitations of this study to note. With regard to protein differential abundance between cases and controls, we were unable to address the critical need to identify proteins that may be biomarkers of key asthma subtypes (e.g., Th2-high versus Th2-low), as such endotyping of cases is not available in the UKB.⁸⁵ While it is possible to stratify patients by eosinophil count as a proxy, this would largely result in the identification of proteins associated with eosinophil counts and not necessarily Th2 status. The same limitation occurs with re-

gard to stratifying by BMI, which confounds asthma status. There are also few comparable studies with which we can test for replication between plasma protein levels and disease status on this scale.

We also acknowledge the limitations of the available genetic and proteomic data, including the fact that we were only powered in the data from UKB-PPP participants of EUR ancestry to identify sufficient pQTLs for genetic prediction of protein levels. Extending these efforts to multi-ancestry and non-EUR ancestry populations is critical not only because assessing variant effects across the allele frequency spectrum can increase the discovery of both pQTLs and asthma risk loci^{2,56,70,78} but also because investigations restricted to a single population may have limited potential for globally equitable clinical translation.⁸⁶ Additionally, while the data generated by the UKB-PPP are some of the most extensive resources available on population-based plasma proteomics to date, we cannot rule out that pQTLs may also impact the levels of plasma proteins not among the 2,923 measured. As a result, there may be additional, unobservable pleiotropic effects confounding associations.

Lastly, selecting variants as IVs for MR is a delicate and imperfect task, especially given the inability to test all IV assumptions directly.⁶⁰ While our approach of using fine-mapped variants as IVs was principled and avoided introducing bias by manually pruning instruments, the challenge of fine-mapping traits with multiple causal variants at a given locus (such as plasma protein levels) may have introduced additional heterogeneity across IV effects and influenced null outcomes. Therefore, while colocalization and MR are valuable tools for further prioritizing putatively causal associations, proteins with negative findings from these analyses should not be interpreted as indications of non-causality without further investigation.

While these limitations highlight opportunities for future work, the collective findings we report here showcase the utility of integrating genetics and plasma proteomics to generate disease-related insights. The application of such approaches in other domains may inform therapeutic development by yielding causal evidence for novel pathologic processes and expanding the ability to derive mechanistic hypotheses from GWAS by leveraging the growing landscape of biobank-scale molecular profiling.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources should be directed to and will be fulfilled by the lead contact, Mark I. McCarthy (mccarthy.mark@gene.com).

Materials availability

This study did not generate new unique reagents.

Data and code availability

- All original code has been deposited at Zenodo (<https://doi.org/10.5281/zenodo.12601581>) and is publicly available as of the date of publication.
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

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AUTHOR CONTRIBUTIONS

L.J.D. and M.I.M. conceived the study. L.J.D., C.B., F.J.I., R.K.P., and A.M. designed analyses and/or generated data. L.J.D., D.C., B.L.Y., A.M., and M.I.M. interpreted the results. L.J.D. and M.I.M. wrote the manuscript. All authors read, revised, and approved the manuscript.

DECLARATION OF INTERESTS

All authors were employees of Genentech, Inc., at the time this work was performed. C.B., D.C., R.K.P., B.L.Y., A.M., and M.I.M. are holders of Roche stock. An immediate family member of M.I.M. is a member of *Cell Genetics*'s advisory board.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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- **METHOD DETAILS**
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 - Differential abundance analysis and characterization of differentially abundant proteins
 - pQTL mapping, finemapping, and generation of GPPLs
 - GPPL associations with asthma and quantitative traits
 - Asthma GWAS in UKB non-PPP population
 - Colocalization
 - Two-sample bi-directional Mendelian randomization
- **QUANTIFICATION AND STATISTICAL ANALYSIS**

SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited data		
UKB-PPP plasma proteomics measurements	UK Biobank	https://www.ukbiobank.ac.uk/enable-your-research
Software and algorithms		
REGENIE v3.2.9	Mbatchou et al. ⁸⁷	rgcgithub.github.io/regenie
PLINK v2.00	Chang et al. ⁸⁸	https://www.cog-genomics.org/plink/2.0/
Ingenuity Pathway Analysis	Krämer et al. ⁸⁹	https://digitalinsights.qiagen.com/IPA
r/coloc v5.2.3	Giambartolomei et al. ⁵⁴ Wallace ⁵³	https://CRAN.R-project.org/package=coloc
r/susieR v.012.41	Zou et al. ⁵⁵	https://CRAN.R-project.org/package=susieR
r/MendelianRandomization	Yavorska et al. ⁹⁰	https://CRAN.R-project.org/package=MendelianRandomization
r/TwoSampleMR	Hemani et al. ⁹¹	https://github.com/MRCIEU/TwoSampleMR
r/MR-PRESSO	Verbanck ⁴⁶	https://github.com/rondolab/MR-PRESSO
Scripts used for analysis in this study	This study	https://doi.org/10.5281/zenodo.12601581

METHOD DETAILS

Definition of asthma cases, controls, and quantitative traits in UKB

Asthma cases were defined as having doctor-diagnosed disease using UKB field 41270 (J45) or self-reported disease using field 20002 (1111). Controls were defined as individuals without a record of asthma or atopic disease defined by field 6152 (value 9) or field 41270 (L40). Both cases and controls were excluded if there was evidence of chronic obstructive pulmonary disease or idiopathic fibrosis using UKB field 41270 (J43, J44, J84) or field 6152.

Cases were defined as incident asthma if the first report of doctor-diagnosed disease asthma (field 42014) was after the date of baseline visit (field 53) when plasma was collected. Asthma age-of-onset was defined by values from fields 22147 and 3786. If values were available for both fields and within 10 years of each other, the value for 22147 was used, otherwise the age-of-onset was set as missing. For asthma-related binary or quantitative traits, the follow UKB fields were used: blood eosinophil counts (30150), blood lymphocyte counts (30120), blood neutrophil counts (30140), and forced expiratory volume in 1-s (FEV₁), predicted percentage (20154), body mass index (21001), tobacco-smoker at time of baseline visit (1239), and cigarette pack-years (20161). Use of asthma-related medication was defined using the previously reported classification of prescriptions reported by Shrine et al.¹³ Participant ancestry group was assigned using PanUKB ancestry definitions in the UKB return dataset 2442.

Differential abundance analysis and characterization of differentially abundant proteins

Plasma protein measurements used in this study were restricted to those for plasma taken at the baseline visit of UKB-PPP participants randomly selected for inclusion in the UKB-PPP and measured in batches 1–6. For proteins measured by multiple Olink assays with strongly correlated measurements (i.e., CXCL8, IDO1, IL6, LMOD1, SCRIB, TNF),¹⁵ values from the Inflammation panel were used. Differential abundance was assessed between controls and asthma cases categorized as doctor-diagnosed asthma and/or self-reported asthma cases, doctor-diagnosed asthma cases only, self-reported asthma cases only, prevalent doctor-diagnosed asthma and/or self-reported asthma cases, incident doctor-diagnosed asthma and/or self-reported asthma, or prevalent doctor-diagnosed asthma cases. Associations between protein levels and asthma status were performed with linear regression using the `lm()` function in R and restricted to individual ancestry groups defined by PanUKB. Protein measurements used were quality controlled NPX values (quality control procedures previously described¹⁵ with outliers excluded using the 1.5*interquartile range method. NPX values were set at the dependent variable with asthma status as the independent variable of interest with age, sex, age*sex, age²*sex, Batch, UKB center, and time between plasma collection and proteomic measurement as covariates (“base” set of covariates). Quantitative covariates were inverse-rank normalized. Significantly associated proteins were defined as those with *p*-values $\leq 1.17 \times 10^{-5}$ (Bonferroni-correction controlling for 2,922 unique proteins tested). The correlation between sets of differentially abundant genes was assessed with the `cor()` function in R on regression betas. Pathway enrichment among significantly differentially abundant proteins was performed with QIAGEN Ingenuity Pathway Analysis using the set of 2,922 measured proteins as the background gene set. Tissue enrichment was assessed using the TissueEnrich R package with the package-defined tissue-specific gene sets generated from GTEx.

pQTL mapping, finemapping, and generation of GPPLs

The pQTL mapping performed by the UKB-PPP consortium has been previously described.¹⁵ Briefly, REGENIE was used to test genome-wide associations with 2,940 proteins (2,922 of which were unique). The level of statistical significance was defined as $p < 1.7 \times 10^{-11}$, corresponding to the standard genome-wide significance level ($p < 5 \times 10^{-8}$) corrected for the number of proteins analyzed ($5 \times 10^{-8}/2940 = 1.7 \times 10^{-11}$). The pQTLs were defined through clumping ± 1 Mb around statistically significant variants, excluding the human leukocyte antigen (HLA) region which is treated as one locus due to complex and extensive LD patterns. Physically overlapping regions were merged into one, deeming the variant with the lowest p -value as the lead variant. We inferred causal variants with FINEMAP across 14,189 pQTLs from the UKB-PPP in 28,008 unrelated individuals of European ancestry. Our motivation to use unrelated individuals is to prevent the possibility of bias when generating genetically predicted protein levels (GPPLs). For FINEMAP we generated summary statistics with a linear model implemented in PLINK2 and allowed for thirty causal variants. GPPLs were generated using fine-mapped variants in the following way:

$$\hat{y}_i = \sum_{j=1}^k g_{ij} \hat{\lambda}_j$$

where $\hat{y}_i$ is the GPPL of the i th individual, g_{ij} is the expected allele dosages of allele B at the j th SNP for the i th individual and $\hat{\lambda}_j$ is the joint effect size estimate of the j th SNP with respect to allele B in the top causal configuration from FINEMAP that includes k variants.

GPPL associations with asthma and quantitative traits

cis-GPPLs and genome-wide GPPLs were selected for association testing if they explained a minimum of 1% of protein level heritability. Associations between GPPLs and binary or quantitative traits were performed with logistic or linear regression, respectively, using data from UKB participants that were not included in the UKB-PPP. Significant thresholds for GPPLs were Bonferroni-corrected for the number of *cis*-GPPLs tested ($n = 1,397$; $p \leq 3.58 \times 10^{-5}$) or genome-wide GPPLs tested ($n = 1,803$; $p \leq 2.77 \times 10^{-5}$).

Asthma GWAS in UKB non-PPP population

We performed genome-wide association testing between doctor-diagnosed or self reported asthma cases ($n = 47,067$) and controls ($n = 254,107$) of European-ancestry which were not selected for the UKB-PPP. Logistic regression was performed using REGENIE with age, sex, 20 principal components, UKB center, and genotyping array as covariates. The set of high-quality genotyped variants tested for association was previously described.¹⁵

Colocalization

Colocalization between pQTLs and asthma risk was performed using coloc and susier⁵³ which allows multiple causal variants at a locus. We defined regions ± 1.5 Mb of the TSS for each protein tested and selected variants tested in both the pQTL mapping and UKB non-PPP asthma GWAS (99.9% shared). We then used `susie_rss()` for finemapping using an in-sample LD-matrix (calculated at each locus independently for pQTLs and asthma risk), followed by `coloc.susie()` on the returned Bayes factors from colocalization. The colocalization prior was calculated based on the number of variants in the tested window. For loci in which no 95% credible set could be identified for asthma risk, we used `coloc.abf()` to test with the single causal variant assumption. The code and output from this analysis are provided in the Zenodo repository (<https://doi.org/10.5281/zenodo.12601581>).

Two-sample bi-directional Mendelian randomization

For Mendelian randomization (MR) using asthma risk as the exposure and protein levels as the outcome, we selected instrumental variables (IVs) from the sentinel variants from the UKB non-PPP asthma GWAS. IVs were removed if they were located in *cis* (within 1Mb) of the protein set as the exposure and were significantly associated with protein levels as they violated the IV assumption or “relevance” and that they are primarily associated with the exposure. For MR with plasma protein levels as the exposure and asthma risk as the outcome, IVs for plasma protein levels were selected from the finemapped pQTLs identified in UKB-PPP population and used to generate GPPLs. We calculated F -statistics to flag IVs that may be weak instruments as well as performed Steiger filtering with the `TwoSampleMR` R package to ensure IVs satisfied the assumption that IVs are more strongly associated with the exposure (plasma protein levels) rather than the outcome (asthma).²² MR model testing for median, inverse-variance weighted (IVW), MR-Egger, and MR-Egger methods was performed with the `MendelianRandomization` R package for proteins with 3 or more IVs including a correction for LD between variants, with testing being limited to the IVW method for proteins with only 1 variant. Heterogeneity was also assessed with Cochran’s Q test statistic. We used the `MR-PRESSO` R package to test for and correct for the effect of outliers.

QUANTIFICATION AND STATISTICAL ANALYSIS

Statistical analysis was performed as described using R 4.2.1.